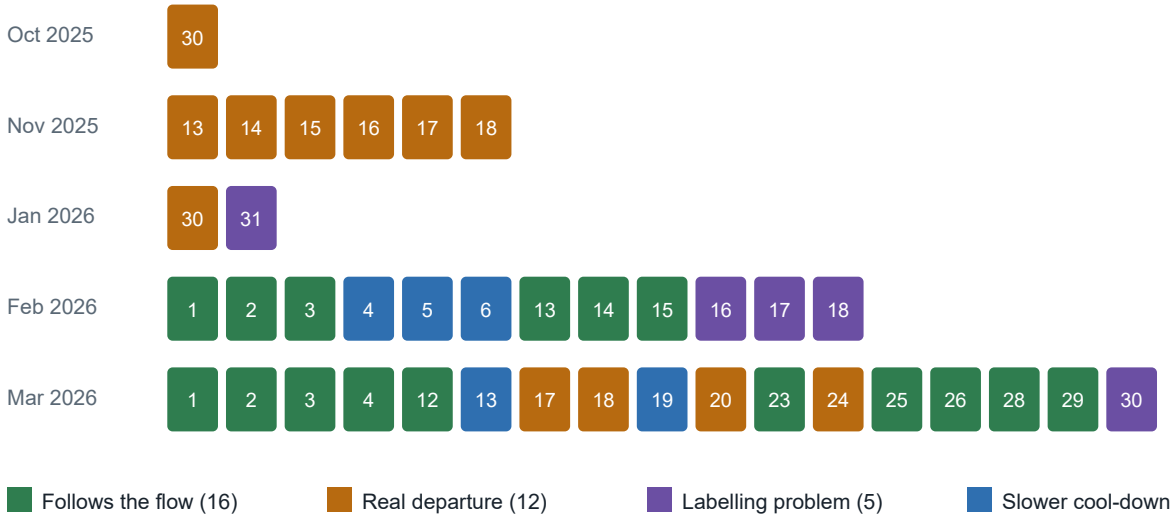
Stage Flow Day Check

Does every day follow the documented flow within each stage?

Contents: 1 Overview · 2 How the check works · 3 Day by day · 4 What to do next

Mostly yes. 16 of the 38 days follow the documented flow with nothing flagged. On the other 22 days the plant did run, but something differs from the document. In most cases the document or the stage labels need fixing; only a few are real departures by the plant.

Why it matters: the operating-modes page is the reference everyone uses to read this plant. If a day does not match it, either the plant did something unusual or the reference is wrong. We need to know which.



Each square is one logged day, coloured by its most important finding. A stage that runs across midnight is checked once, as a whole, and its result shows on every day it covers.

Result	Days	What it means
Follows the flow	16	Every check passed: right order of stages, temperatures and speeds within the documented ranges.
Real departure	12	The plant did something the flow does not describe, mostly heat-ups stopped before the stack was ready.
Labelling problem	5	The stage-naming program still calls a hot stack "Cold / idle", against the rule that Shut-down lasts until below 50 °C.
Slower cool-down	5	The stack cooled at about −10 to −27 °C/h; the document says −45 to −75. Seen on 6 of 10 shut-downs.

If you remember one thing
The plant follows its documented flow on 16 of 38 days. Most mismatches point to fixes in the document or the labelling program, not to plant faults.

2 · HOW THE CHECK WORKS

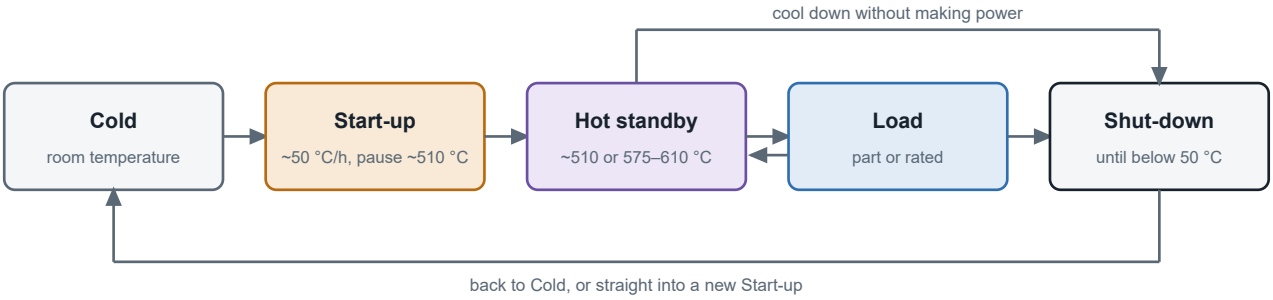
Each stage is held up against the document's own description of it

A program reads every day's sensor readings and names the stage the plant was in each minute. A second program then compares each stage, from start to finish, with what the operating-modes document says should happen in that stage.



Steps 1 and 2 use the stage-naming program that already exists. Steps 3 to 5 are the new check. Joining the days first means a three-day hold at ~510 °C is checked as one hold, not three pieces.

The flow every day is compared with



The plant may stop at any stage, but it should only move along these arrows. Load can drop back to Hot standby and climb again.

What is checked inside each stage

Stage	What should happen (from the document)	Allowed range
All stages	Stages follow the arrows in the diagram	exact
All stages	Hottest sensor stays below the maximum	≤ 720 °C (limit)
Start-up	Starts near room temperature when coming from Cold	10–40 °C
Start-up	First heat-up speed	34–56 °C/h
Start-up	Never heats faster than the maker's limit	≤ 120 °C/h (limit)
Start-up	Short pause at ~510 °C, shorter than 6 hours	490–530 °C, < 6 h
Start-up	Ends in Hot standby	exact
Hot standby	Low hold: parked at ~510 °C for at least 6 hours	480–540 °C, ≥ 6 h
Hot standby	Ready level, before load	575–610 °C
Hot standby	Climb from the low hold to the ready level	8–27 °C/h
Load	Current raised per string, per minute	0.3–0.9 A/min

Stage	What should happen (from the document)	Allowed range
Load	Never raised faster than the maker's limit	≤ 1 A/min (limit)
Load	Current lowered per string, per minute	−1 to −3 A/min
Shut-down	Cooling with the air blower on, 600 → 300 °C	−45 to −75 °C/h
Shut-down	Cooling with the air blower on, 300 → 100 °C	−27 to −57 °C/h
Shut-down	Cooling on its own after the blower stops	−1 to −2 °C/h
Shut-down	Lasts until below 50 °C, or a new Start-up begins	< 50 °C (your rule)

The measured ranges were first read off these same readings, so a value still passes if it is just outside, by up to 20 % of the range's width. The maker's limits and the rules you set get no slack at all.

To run it again (in the project folder)

```
python data_analysis/flow_check.py    all days
python data_analysis/flow_check.py 20260319    one or more days
Results: data_analysis/flow_check/per_day.csv, checks.csv, episodes.csv
```

> **If you remember one thing**
> Every stage visit is compared, from start to finish, with the document's own description of that stage. A day passes only if all of its stage visits pass.

All 38 days: the stages, and what did not match

Each line shows the day's stages in order, then any finding, with the measured value beside the document's value. A finding about a stage that spans several days (for example the five-day cool-down in November) shows on each of those days.

Day	Stages in order	Finding (measured · document)
2025-10-30	Cold / idle → Start-up → Cold / idle	**Real departure**: Start-up began warm, not from room temperature · 139.5 · 10 … 40 **Real departure**: Start-up stopped before reaching Hot standby · Cold / idle · Hot standby **Real departure**: Stage order not in the diagram · Start-up -> Cold / idle · in §2 diagram
2025-11-13	Cold / idle → Start-up → Shut-down	**Real departure**: Mid-way hold at a different temperature · 471.7 · 490 … 530 **Real departure**: Start-up stopped before reaching Hot standby · Shut-down · Hot standby **Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **To confirm**: One heat-up reading above the 120 °C/h limit · 148.5 · ≤ 120.0 **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27
2025-11-14	Shut-down	**Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27
2025-11-15	Shut-down	**Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27
2025-11-16	Shut-down	**Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27
2025-11-17	Shut-down	**Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27
2025-11-18	Shut-down → Cold / idle	**Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -8.8 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -5.4 · -57 … -27

Day	Stages in order	Finding (measured · document)
2026-01-30	Cold / idle → Start-up → Shut-down → Cold / idle	**Real departure**: Start-up stopped before reaching Hot standby · Shut-down · Hot standby **Real departure**: Stage order not in the diagram · Start-up -> Shut-down · in §2 diagram **Labelling problem**: Shut-down label ended while the stack was still hot · 197.4 · -50 … 50.0 **Labelling problem**: "Cold / idle" label on a stack that was still hot · 197.3 · -50 … 50.0
2026-01-31	Cold / idle → Start-up → Hot standby	**Labelling problem**: "Cold / idle" label on a stack that was still hot · 197.3 · -50 … 50.0 **Labelling problem**: Start-up began warm, not from room temperature · 176.0 · 10 … 40
2026-02-01	Hot standby	**Follows the flow** · 5 checks passed
2026-02-02	Hot standby	**Follows the flow** · 5 checks passed
2026-02-03	Hot standby	**Follows the flow** · 5 checks passed
2026-02-04	Hot standby → Shut-down	**Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -21.9 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -8.3 · -57 … -27
2026-02-05	Shut-down	**Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -21.9 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -8.3 · -57 … -27
2026-02-06	Shut-down → Cold / idle → Start-up	**Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -21.9 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -8.3 · -57 … -27
2026-02-13	Cold / idle → Start-up	**Follows the flow** · 8 checks passed
2026-02-14	Start-up → Hot standby	**Follows the flow** · 11 checks passed
2026-02-15	Hot standby	**Follows the flow** · 4 checks passed
2026-02-16	Hot standby → Shut-down	**Labelling problem**: Shut-down label ended while the stack was still hot · 210.0 · -50 … 50.0 **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -19.6 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -8.8 · -57 … -27
2026-02-17	Shut-down → Cold / idle → Start-up	**Labelling problem**: Shut-down label ended while the stack was still hot · 210.0 · -50 … 50.0 **Labelling problem**: "Cold / idle" label on a stack that was still hot · 210.0 · -50 … 50.0 **Labelling problem**: Start-up began warm, not from room temperature · 191.7 · 10 … 40 **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -19.6 · -75 … -45 **Slower cool-down**: Cool-down slower than the document says (300→100 °C) · -8.8 · -57 … -27
2026-02-18	Start-up → Hot standby → Shut-down	**Labelling problem**: Start-up began warm, not from room temperature · 191.7 · 10 … 40 **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -27.0 · -75 … -45
2026-03-01	Cold / idle → Start-up → Hot standby	**Follows the flow** · 14 checks passed
2026-03-02	Hot standby	**Follows the flow** · 6 checks passed
2026-03-03	Hot standby	**Follows the flow** · 6 checks passed
2026-03-04	Hot standby → Shut-down	**Follows the flow** · 9 checks passed

Day	Stages in order	Finding (measured · document)
2026-03-12	Cold / idle → Start-up	**Follows the flow** · 9 checks passed
2026-03-13	Hot standby → Part load → Hot standby → Part load → Shut-down	**Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -23.4 · -75 … -45
2026-03-17	Cold / idle → Start-up → Cold / idle	**Real departure**: Start-up began warm, not from room temperature · 57.7 · 10 … 40 **Real departure**: Start-up stopped before reaching Hot standby · Cold / idle · Hot standby **Real departure**: Stage order not in the diagram · Start-up -> Cold / idle · in §2 diagram
2026-03-18	Cold / idle → Start-up → Hot standby	**Real departure**: Stage order not in the diagram · Start-up -> Cold / idle · in §2 diagram **Real departure**: Start-up began warm, not from room temperature · 168.5 · 10 … 40 **Real departure**: Heat-up speed outside the usual range · 61.4 · 34 … 56
2026-03-19	Hot standby → Part load → Hot standby → Part load → Shut-down	**Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -11.6 · -75 … -45
2026-03-20	Shut-down → Start-up → Hot standby → Shut-down	**Real departure**: Heat-up speed outside the usual range · 27.6 · 34 … 56 **Slower cool-down**: Cool-down slower than the document says (600→300 °C) · -11.6 · -75 … -45
2026-03-23	Start-up → Hot standby	**Follows the flow** · 11 checks passed
2026-03-24	Hot standby → Part load → Hot standby → Part load → Hot standby	**Real departure**: Standby temperature above the usual band · 628.9 · 575 … 610
2026-03-25	Hot standby → Part load → Hot standby → Part load → Hot standby → Part load → Shut-down	**Follows the flow** · 30 checks passed
2026-03-26	Shut-down → Cold / idle	**Follows the flow** · 8 checks passed
2026-03-28	Cold / idle → Start-up	**Follows the flow** · 8 checks passed
2026-03-29	Hot standby	**Follows the flow** · 6 checks passed
2026-03-30	Hot standby → Part load → Hot standby → Shut-down → Cold / idle	**Labelling problem**: Shut-down label ended while the stack was still hot · 264.3 · -50 … 50.0 **Labelling problem**: "Cold / idle" label on a stack that was still hot · 264.3 · -50 … 50.0

If you remember one thing

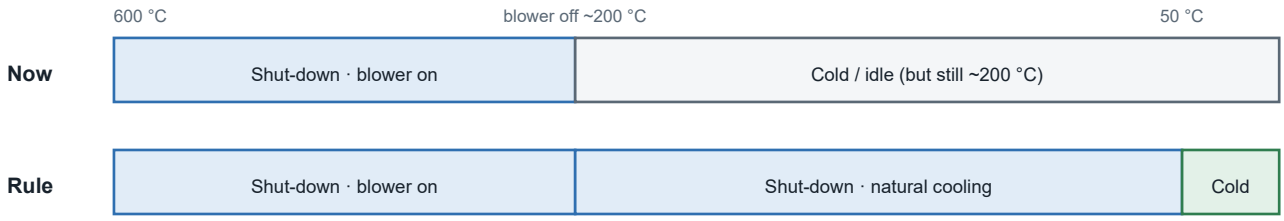
The days that match cluster in the February and March holds and the late-March runs. Most flagged days belong to a handful of stopped heat-ups and slow cool-downs.

Four follow-ups: a program fix, a document change and two questions

Much of the plant already runs as documented. Every load ramp stayed under the maker's limit of 1 A per minute (highest: 0.92). Every hold sat at its documented temperature. No sensor went above 720 °C.

1 · Fix the labelling program: a hot stack is still called "Cold / idle"

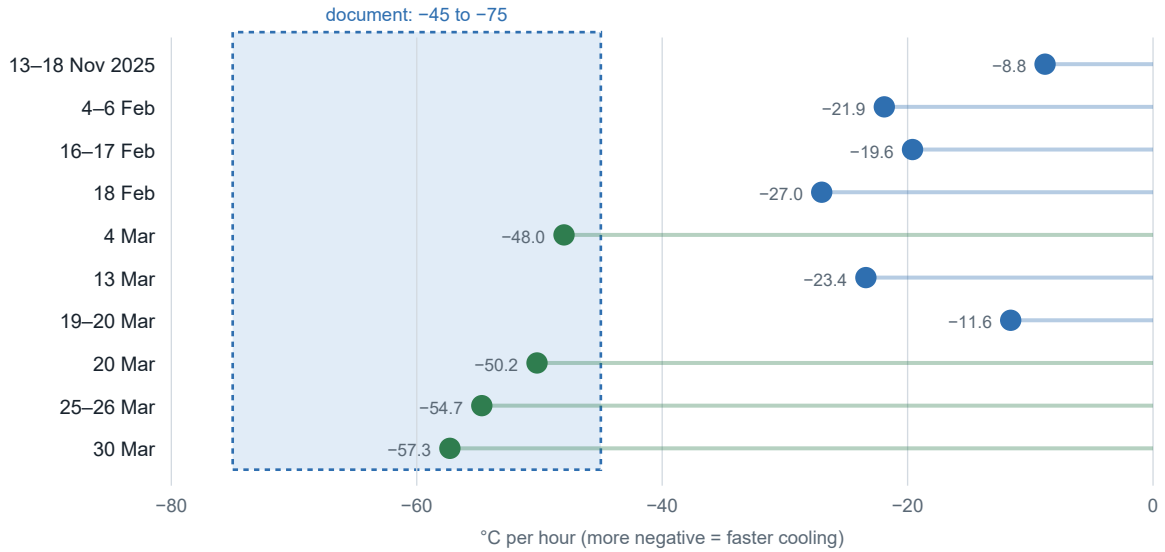
On 30 Jan, 16–17 Feb and 30 Mar the air blower was switched off while the stack was still at 197–264 °C. The program then calls the stage "Cold / idle". Your rule says Shut-down continues until the stack is below 50 °C. Because of this, the next heat-up on 31 Jan and 17–18 Feb also looks as if it started warm "from Cold". Where: data_analysis/operating_states.py (classify_row / apply_context).



The cooling stays the same. Only the name of the stretch after the blower stops changes.

2 · Update the document: the plant cools down at two speeds

Of the 10 shut-downs that pass 600 → 300 °C with the blower on, 4 cool as fast as the document says. The other 6 cool two to five times more slowly. Before changing §3.3, ask the plant team whether the slow cool-downs were a deliberate gentler setting.



Each dot is one shut-down; the shaded band is the document's range. Speed is timed from the last moment at 600 °C (or the highest point) to reaching 300 °C.

3 · Ask the plant team: six times the plant did something the flow does not show

When	What happened	Document says
30 Oct 2025	First file starts mid heat-up at 140 °C; the heat-up stops and the stack settles back	Start-up ends in Hot standby
13–18 Nov 2025	Heat-up pauses at 472 °C, then the stack cools for five days without reaching Hot standby	Pause at ~510 °C, then on to Hot standby
30 Jan 2026	Heat-up stops and goes straight into a cool-down	Start-up ends in Hot standby
17–18 Mar 2026	Heat-up from 58 °C stops; restarts next day from 169 °C at 61 °C/h	34–56 °C/h (limit 120)
20 Mar 2026	Heat-up at 28 °C/h, slower than usual	34–56 °C/h
24 Mar 2026	Standby held at 629 °C between load runs	575–610 °C (max 720)

If stopping a heat-up part-way is a normal operator choice, the transition diagram (§2) should get a "stopped start-up" arrow.

4 · Confirm one reading: one heat-up speed above the limit is probably a logger glitch

On 13 Nov 2025 the hottest sensor shows 148 °C per hour, above the 120 °C/h limit. Just before, the readings sit frozen at 217 °C for about 15 minutes, then jump 69 °C in five minutes. That looks like the logger repeating stale values. The document records 108 °C/h for this heat-up. Check the logger before treating it as a breach.

> **If you remember one thing**
> Fix the labelling program first: it is fully in our hands, and on its own it clears 31 Jan and 30 Mar. The other findings need an answer from the plant team before the document changes.